



DATABASE SYSTEMS

(CS-151)

Topic: Database Environment

Course Instructor: Munazza



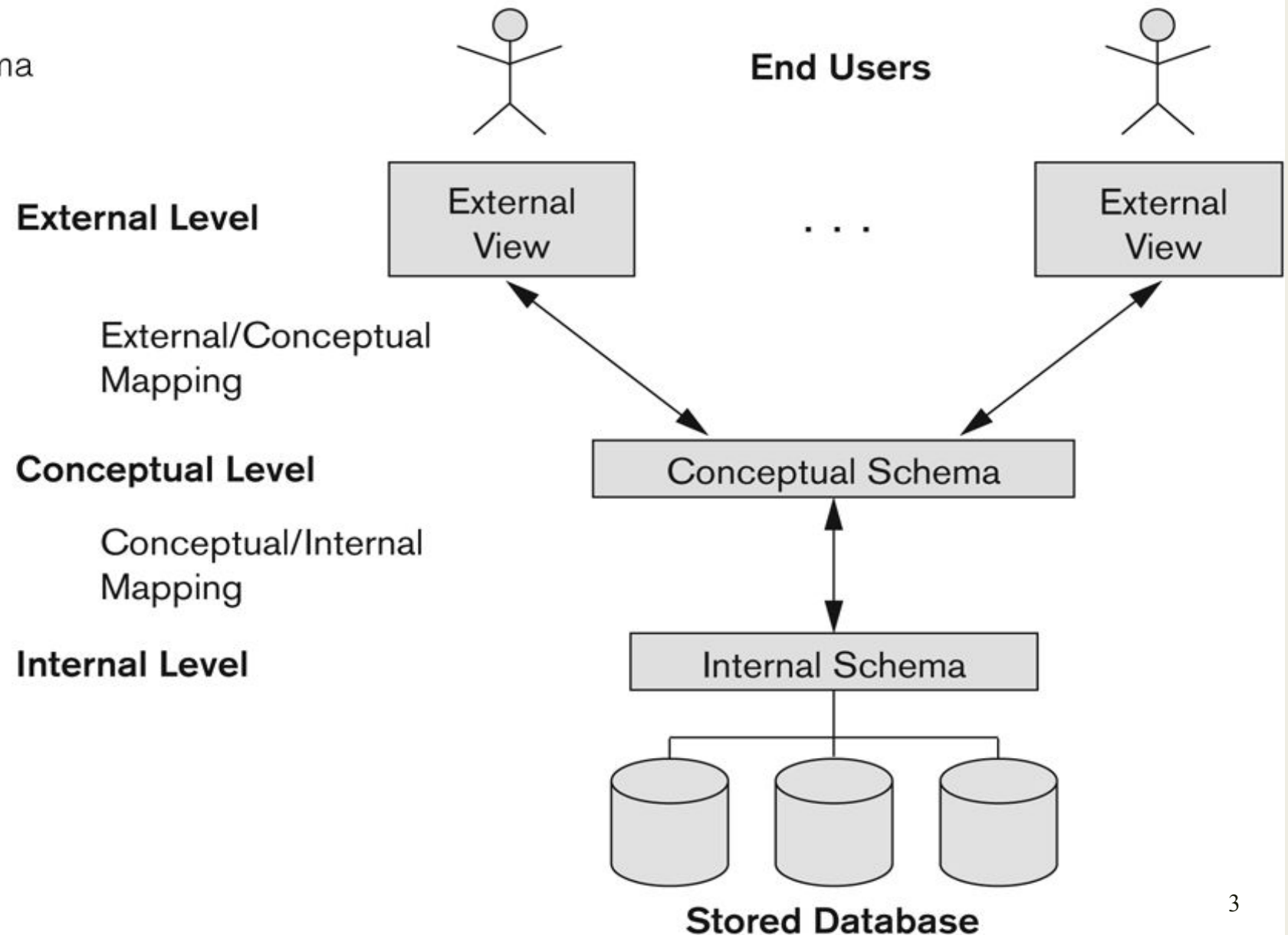
Lecture Overview

- Database Architecture
- Data Independence
- Different aspects of DBMS

The Three-Schema Architecture

Figure 2.2

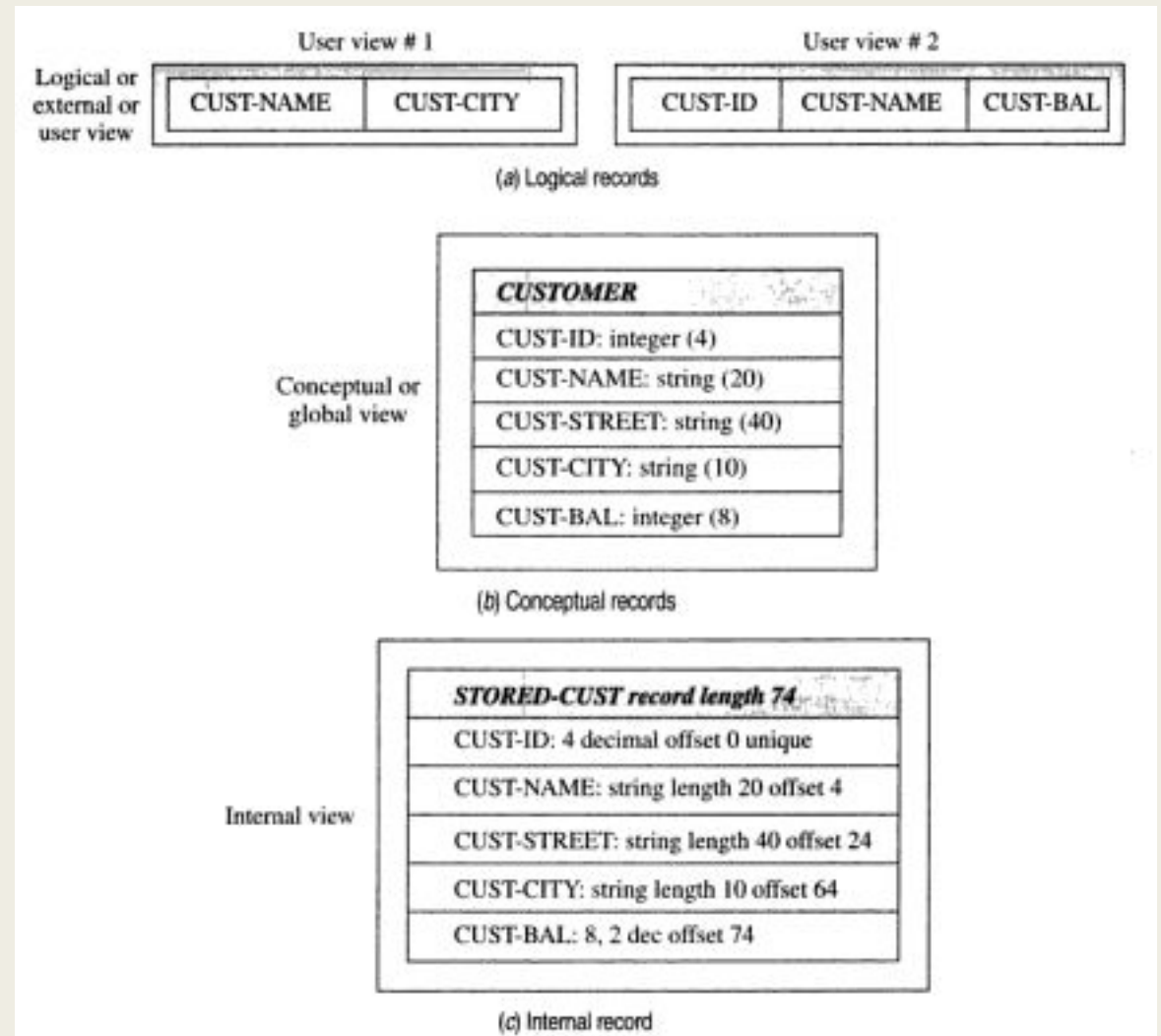
The three-schema architecture.



Three-Level ANSI-SPARC Database Architecture

- ANSI-SPARC three-tier architecture consists of three levels

- *Internal level*
- *Conceptual level*
- *External level*



External Level

- It is the view how the user views the database.
- The external level consists of several different external views of the database.
- Highest level of data abstraction
- Describes part of the database that is relevant to the user
- Different views may have different representation of the same data.
- In the external view only that entities, attributes, and relationships are included that the user wants.
- Frontend Designer

External Level

- Example: one may view the date as d/m/y while other view the date as y/d/m
- Some view may include derived or calculated data
- For Example,
 - *One user may view name in the form (firstname, lastname), while another may view as (lastname, firstname)*

Conceptual Level

- Middle level in three tier architecture
- Contains the logical **structure** of the entire database as seen by the DBA, also called logical schema
- Describes what **data** is stored in the database and **relationships** among them
- DB Designer
- Conceptual level is concerned with the following activities:
 - *All entities, their attributes and their relationships*
 - *Constraint on the data*
 - *Semantic information about the data*
 - *Checks to retain data consistency and integrity*
 - *Security information*

Internal Level

- Lowest level of abstraction of database
- How data will be stored in the database
- Describes the data structure, file structures, and access methods to be used by the database
- The internal schema contains the definition of the stored record, the method of representing the data fields, indexing and hashing schemes and access methods used
- Summarizes how the relations described in the conceptual schema are actually stored on secondary storage devices such as disks and tapes
- DBA

Internal Level

- Internal level is concerned with the following activities
 - *Storage space allocation for data and storage*
 - *Record description for storage with stored sizes of data items*
 - *Record placement*
 - *Data compression and data encryption techniques*

Advantages of Three-tier Architecture

- It allows to separate the physical representation of the data from the user's view of data
- Each user is able to access the same data but have different customized view as their requirements
- Not all the users have to access the data in database at all database levels
- Internal structure of the database is unaffected by changes to the physical storage organization
- DBA can change the database storage requirements and also the conceptual structure of the database without affecting user requirements

Data Independence

- Characteristics of the database system to change the schema at one level without having to change at the next higher level
- Application programs do not depend on any particular physical representation or access technique
- Two types of data independence
 - *Physical Data Independence*
 - *Logical Data Independence*

Physical Data Independence

- Immunity of the conceptual schema to changes in the internal schema is referred to as physical data independence
- Change in the internal schema without affecting the conceptual or external schemas.
- Possible changes to the internal schema are: using different file organization, storage structures, using different storage devices, modifying indexing or hashing algorithms
- DB shift from one area to another, does not affect conceptual(no change in tables)

Logical Data Independence

- Immunity of the external schema to changes in the conceptual schema is referred to as logical data independence
- Changes to the conceptual schema must be possible without changing existing external schemas
- Possible changes to the conceptual schema are: addition and deletion of entities, addition and deletion of attributes, addition and deletion of relationships
- Changes by one user do not affect other, the app run as it is

Mapping

- Mapping is defined as the mechanism through which data at one level is related to the changed format at the other level is known as mapping
- DBMS is responsible for this mapping between internal, conceptual and external schemas.
 - *User's request specifies at external schema level must be transformed into request at conceptual schema level*
 - *The transformed request at the conceptual schema level should be further transformed at the internal schema level for final processing of the data in the stored database as per user's request*
 - *The final result from the processed data as per user's request must be reformatted to satisfy the user's external view*

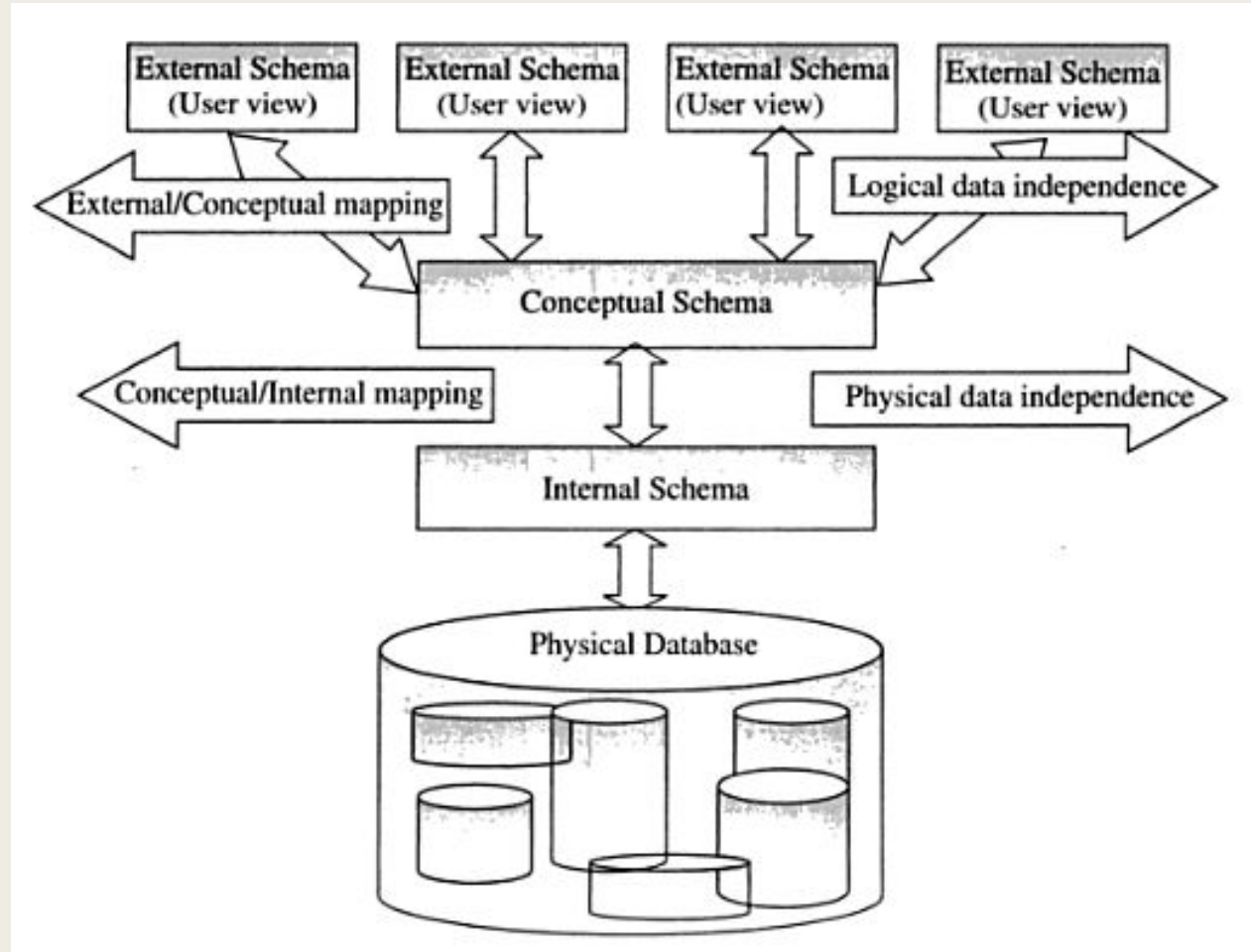
Conceptual/Internal Mapping

- Defines the correspondence between conceptual view and the stored database
- Specifies how conceptual records and fields are presented at the internal level
- Enables the DBMS to find the actual record or combination of records in physical storage that constitutes the logical record in the conceptual schema
- A change to the storage structure definition means that the conceptual/internal mapping must be changed accordingly, so that the conceptual schema may remain invariant, achieving physical data independence.

External/Conceptual Mapping

- Defines the correspondence between a particular external view and the conceptual view
- It gives the correspondence among the records and relationships of the external and conceptual view
- Enables DBMS to map names in the user's view on to the relevant part of the conceptual schema
- Several mappings between external and conceptual level
- A change to the conceptual definition means that the conceptual/external mapping must be changed accordingly, so that the external schema may remain invariant, achieving logical data independence.

Mapping & Dependence of Three Tier Architecture



Database System Environment

- Database system refers to an organization of components that define and regulate the collection, storage, management and use of data within a database environment.
- It consists of four main parts
 - *Data*
 - *Hardware*
 - *Software*
 - *Users (People)*

Database Language

- DBMS must provide appropriate language and interfaces for each category of users to express database queries and updates
- Following languages are used to specify database schema
 - *Data Definition Language*
 - *Storage Definition Language*
 - *View Definition Language*
 - *Data Manipulation Language*
 - *Fourth- Generation Language*

Data Definition Language (DDL)

- Used to specify a database conceptual schema using set of definitions
- Supports definition and declaration of database objects
- Allows DBA or user to describe and name the entities, attributes and the relationships required for the application, together with any associated integrity and security constraints
- For example, to build a new table using SQL syntax, to CREATE statement/command is used, followed by parameters for the table name and column definitions.

Data Manipulation Language (DML)

- Mechanism that provides a set of operations to support the basic data manipulation operations on the data held in the database
- Helps in communicating with the DBMS
- DML provides following functional access
 - *Retrieve data and/or records from the database*
 - *Add records to the database files*
 - *Delete records from the database file*
 - *Retrieve records sequentially in the key sequence*
 - *Retrieve records in the physically recorded sequence*
 - *Rewrite records that have been updated*
 - *Modify data and/or records in the database file*

Data Storage Definition Language (DSDL)

- Used to specify the internal schema in the database
- It defines
 - *Mapping between conceptual schema and internal schema*
 - *Storage structure*
 - *Access methods*
- Specify set of statements which define the implementation details of database schemas, which are usually hidden from the users

View Definition Language (VDL)

- Used to specify user's view and their mappings to the conceptual schema
- Two views of data are:
 - *Logical View*
 - *Physical View*

Fourth-Generation Language (4GL)

- Compact, efficient and non procedural programming language
- Used to improve the productivity of the DBMS
- It has following inbuilt components:
 - *Query languages*
 - *Report generators*
 - *Spreadsheets*
 - *Database languages*
 - *Application generator to define operations such as insert, retrieve and update data from the database to build applications*
 - *High level languages to generate application program*
- Examples: SQL and QBE(Query by Example)